
BY 523P: Systems Biology Lab

Course Code	BY 523P
Course Name	Systems Biology Lab
L-T-P-C	0-0-2-1
Prerequisites	None
Intended for	M. Tech. Biotechnology
Distribution	Core Lab for M. Tech. Biotechnology
Approval	9th Senate (May 2016)

Course Contents

Lab sessions for **Introduction to “OMICS” and Systems Analysis** course:

- Basic Insilico analysis of NGS data of genomes/Metagenomes
- Differential gene expression analysis of transcriptome data
- 16S rRNA based phylogenetic profiling
- Introduction to Gene Ontology, KEGG, EcoCyc databases
- Automated pathway mapping and annotation of proteins
- Extraction of reaction and metabolite information
- Metabolic network reconstruction
- Genome scale model analysis
- Insilico Gene dispensability analysis

Lab sessions for **Metagenomics, and Next Generation Sequencing Technologies** course:

- Next Generation Sequencing application for Amplicon sequencing
- Next Generation Sequencing application for Transcriptome sequencing
- Next Generation Sequencing application for Metagenomic sequencing
- Basic Insilico analysis of NGS data of genomes/Metagenomes
- Differential gene expression analysis of transcriptome data
- 16S rRNA based phylogenetic profiling

Lab sessions for **Metabolic Systems Biology** course:

- Metabolite profiling experiment – NMR/MS based
- Metabolomics experiment untargeted/targeted
- Statistical analysis of the metabolomics data generated
- Measuring major biomass components (for constraints) of a typical cell – I
- Measuring major biomass components (for constraints) of a typical cell – II
- Isotopomer measurements using Mass spectroscopy and their analysis
- ¹³C based/constraint based flux analysis of central metabolic network-I
- ¹³C based/constraint based flux analysis of central metabolic network-II
- Making sense of the generated flux map.

Lab sessions for **Proteomics** course:

- Proteomics experiments